

Christoph R. Landolt

Doctoral Researcher at CISPA Helmholtz Center for Information Security, working on adversarial AI, multi-agent learning, and game theory for cybersecurity.

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🐙 github.com/clandolt 🎓 Google Scholar 🇨🇭 Swiss citizen 📅 02.11.1992
🗣 German (native) · English (fluent) · French (fluent)



EXPERIENCE

ELLIS PhD Student

May 2025 – Present

CISPA Helmholtz Center for Information Security · Saarbrücken

- Research on multi-agent & equilibrium learning, reinforcement learning, and game theory for cybersecurity.
- **Advisors:** Mario Fritz (CISPA), Florian Tramèr (ETH Zürich).
- **Teaching:** Teaching Assistant for *Machine Learning in Cybersecurity*.

CYD Research Fellow

Oct 2024 – May 2025

Cyber-Defence Campus (armasuisse Science & Technology) · Lausanne

- **Thesis:** "Training Offensive Agents with Multi-Agent Reinforcement Learning (MARL)."
- **Advisors:** Julian Jang-Jaccard (Cyber-Defence Campus), Christoph Würsch (OST).
- Results published at ICMCIS 2025 (arXiv:2505.19837); follow-up published at CyCon 2026 (Proceedings).

Research Assistant (ML)

Oct 2022 – Oct 2024

Institute for Computational Engineering (ICE), OST – Eastern Switzerland University of Applied Sciences · Buchs SG

- Led applied research in deep learning (ConvRNNs, ResNet) and recommender systems, transferring results into industry applications.
- Designed the curriculum for the CAS in AI-Driven Cybersecurity and Strategic Defence (game theory, adversarial AI).
- Mentored students as a Study Tutor for 30 students in Computer Science.

AWARDS

- **CYD Master Thesis Fellowship**, EPFL & Cyber-Defence Campus – Selected through a highly competitive, merit-based process to receive a 6-month research grant and living allowance for outstanding academic excellence in cyber-defence and data science.

SKILLS

DL Frameworks: PyTorch, JAX/Flax, TensorFlow

ML & RL: Multi-Agent RL, game theory, reinforcement learning, deep learning

Programming: Python, PowerShell, Bash, SQL, C#

Security: network access control (PNAC), vulnerability assessment

Systems: server administration, network administration

Staff Officer (Captain, OF-2)

Oct 2013 – May 2025

Cyber Command, Swiss Armed Forces (compulsory military service) · Switzerland

- 2022–2025: Staff Officer (Quartermaster) in the specialized cyber staff.
- 2018–2021: Finance, accommodation, food, logistics, and administration for a battalion of 1,000–1,500 soldiers.
- 2013–2017: Same scope for a unit of 200–300 soldiers as Quartermaster Sergeant.

System Administrator

Sep 2017 – Jan 2018

Abraxas Informatik AG · Zürich

- Automated workflows for Endpoint and Configuration Management.
- Collaborated on implementing a port-based network access control (PNAC) solution, improving network security and access control for clients.

System Specialist

Aug 2013 – Apr 2016

Abraxas Informatik AG · St. Gallen / Zürich

- Designed, implemented, and maintained internal and external infrastructures and services, including directory services, messaging, unified communications, and monitoring.
- Evaluated and rolled out new technologies for automation and standardization.
- Led sub-projects within large-scale infrastructure programs.
- Trained and mentored third- and fourth-year apprentices.

Apprentice in System Technology

Aug 2009 – Aug 2013

Abraxas Informatik AG · St. Gallen

- Foundations in security best practices: access control, vulnerability assessments, patch management.
- Foundations in software engineering and debugging, programming in PowerShell, SQL, and C#.

PROJECTS

Tutorial Notebooks – Machine Learning in Cybersecurity

- Exercise notebooks on ML applications in cybersecurity – attacks, defenses, and AI for CTF challenges – for the CISPA graduate course (website).

Consensus-Driven Multi-Agent Language Model Framework

- Researched alignment strategies among multiple LLM agents collaborating to solve coding problems.

EDUCATION

PhD in Computer Science (Candidate)

May 2025 – Present

Universität des Saarlandes · Saarbrücken
• Research at CISPA Helmholtz Center.

MSc Engineering, Specialization in Data Science

Oct 2022 – Mar 2025

OST – Eastern Switzerland University of Applied Sciences ·
Buchs SG
• **Advisors:** Christoph Wüsch, Klaus Frick.
• **CYD Fellowship Mentor:** Julian Jang-Jaccard.

Advanced Course on Data Science & Machine Learning

June 2024

ICAS – The Interdisciplinary Centre of Advanced Studies ·
Tuscany, Italy
• Summer school on Deep Learning, Generative AI, and Rein-
forcement Learning (8 ECTS for PhD students).

BSc FHO Systems Engineering, Specialization in Computer Science

Aug 2019 – Sep 2022

OST – Eastern Switzerland University of Applied Sciences · St.
Gallen · GPA 5.27/6
• **Advisors:** René Pawlitzek, Klaus Frick.
• Thesis on anomaly detection in web traffic with machine learn-
ing.

Maturity (Passerelle)

Aug 2016 – Aug 2017

Interstaatliche Maturitätsschule für Erwachsene St. Gallen · St.
Gallen · GPA 5.2/6
• Additional exam: "Passerelle Berufsmaturität – universitäre
Hochschule."

Technical Vocational Maturity (Part-time)

Aug 2014 – Aug 2016

Berufs- und Weiterbildungszentrum Wil-Uzwil · Wil · GPA 5.2/6

Information Technologist, Federal Diploma of VET

Aug 2009 – Aug 2013

St. Gallen Centre of Vocational Education and Training (GBS) · St.
Gallen · GPA 5.1/6

PUBLICATIONS

- [1] **Landolt, C. R.**, Lorenz, T., Kwiatkowska, M., Fritz, M. "The Oracle's Gambit: A Game-Theoretic Framework for Responsible AI Release." *arXiv preprint*, 2026. arXiv:2607.05442
- [2] **Landolt, C. R.**, Jang-Jaccard, J., Mulder, V., Meier, R., Wüsch, C., Fritz, M. "Enhancing Cyber Attack Autonomy Through Multi-Agent Reinforcement Learning." *International Conference on Cyber Conflict (CyCon)*, NATO CCDCOE, 2026. Proceedings
- [3] Ammann, L., Ott, S., **Landolt, C. R.**, Lehmann, M. P. "Securing RAG: A Risk Assessment and Mitigation Framework." *IEEE Swiss Conference on Data Science (SDS)*, 2025. arXiv:2505.08728
- [4] **Landolt, C. R.**, Wüsch, C., Meier, R., Mermoud, A., Jang-Jaccard, J. "Multi-Agent Reinforcement Learning in Cybersecurity: From Fundamentals to Applications." *International Conference on Military Communication and Information Systems (ICMCIS)*, 2025. arXiv:2505.19837

TALKS & MEDIA

05/2026 Conference talk at CyCon 2026 (NATO CCDCOE) on "AI-Driven Autonomy in Cyber Conflict: Offense, Defence and Agentic Systems" (Video).
03/2026 Radio interview with Deutschlandfunk, *Forschung Aktuell*, on generative AI and software security (Listen).
06/2025 Talk at the Cyber Alp Retreat, Davos, on "Multi-Agent Reinforcement Learning for Cybersecurity" (Slides).

EXECUTIVE & LEADERSHIP EDUCATION

Tactical Leadership Course (General Staff School)

2024

Armed Forces College (AFC) · Luzern

- Führungslehrgang Grosser Verband (FLG Gr Vb) – executive training for operational planning and decision-making at the brigade/division level.

Leadership Bootcamp (Academy Certificate)

2024 – 2025

HWZ University of Applied Sciences in Business Administration ·
Zürich

- Professional training in agile leadership, communication strategies, and conflict management.

Tactical Leadership Course (Battalion Level)

2019

Armed Forces College (AFC) · Luzern

- Führungslehrgang Truppenkörper (FLG Trp Kö) – Command and Staff Officer School; leadership training for staff officers on battalion staffs.

Quartermaster Officer Candidate (Logistics Officer School)

Mar 2018 – Oct 2018

Swiss Armed Forces – Logistik Offiziersschule (Log OS) · Bern

- Log OS 40-1: military logistics & leadership education – command & staff techniques, operational planning, and leadership in normal and crisis situations.

VOLUNTEERING

Associate Editor

Jul – Nov 2024

16th International Conference on Social Robotics + AI · Odense,
Denmark

- Managed a team of sub-reviewers and reviewed AI submissions.

Head of Finance

Aug 2020 – Aug 2022

Student Organization, Buchs Campus · Buchs

- Project lead for the diploma magazine and the graduation/campus festival.
- Co-managed the merger of the student organizations across the Buchs, St. Gallen, and Rapperswil campuses.

Treasurer & Chairman, St. Gallen Local Group

2019 – Present

Swiss Fourier Association, Eastern Switzerland Section · St. Gallen

- Budgeting, resource planning, and technical events for a 300-member section.